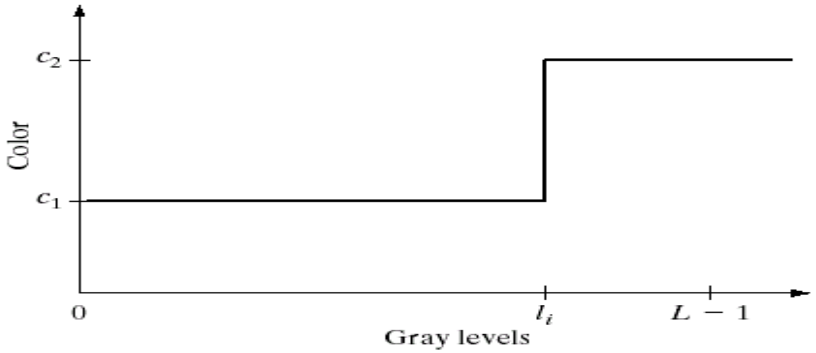
Lecture - 14

Chapter 6

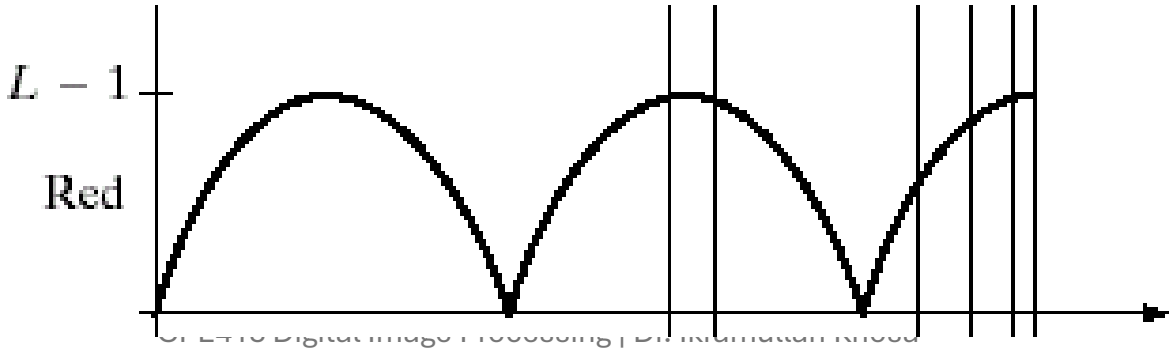
COLOR IMAGE PROCESSING

Pseudo-color Image Processing

- Intensity slicing: piecewise linear transformation



- General Gray level to color transformation



Gray level to Color transformation

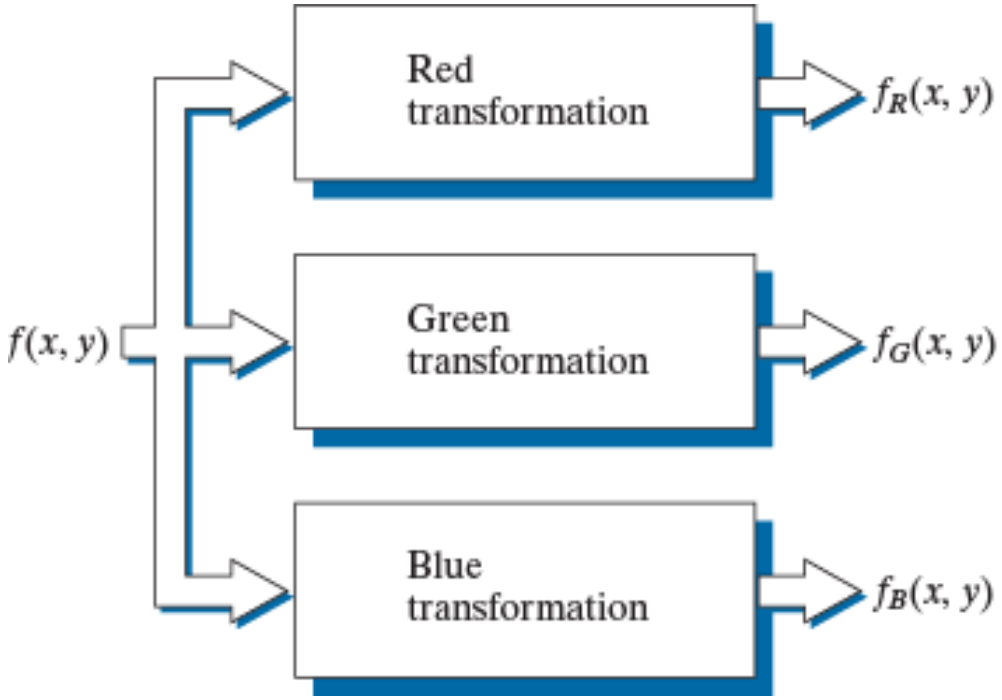
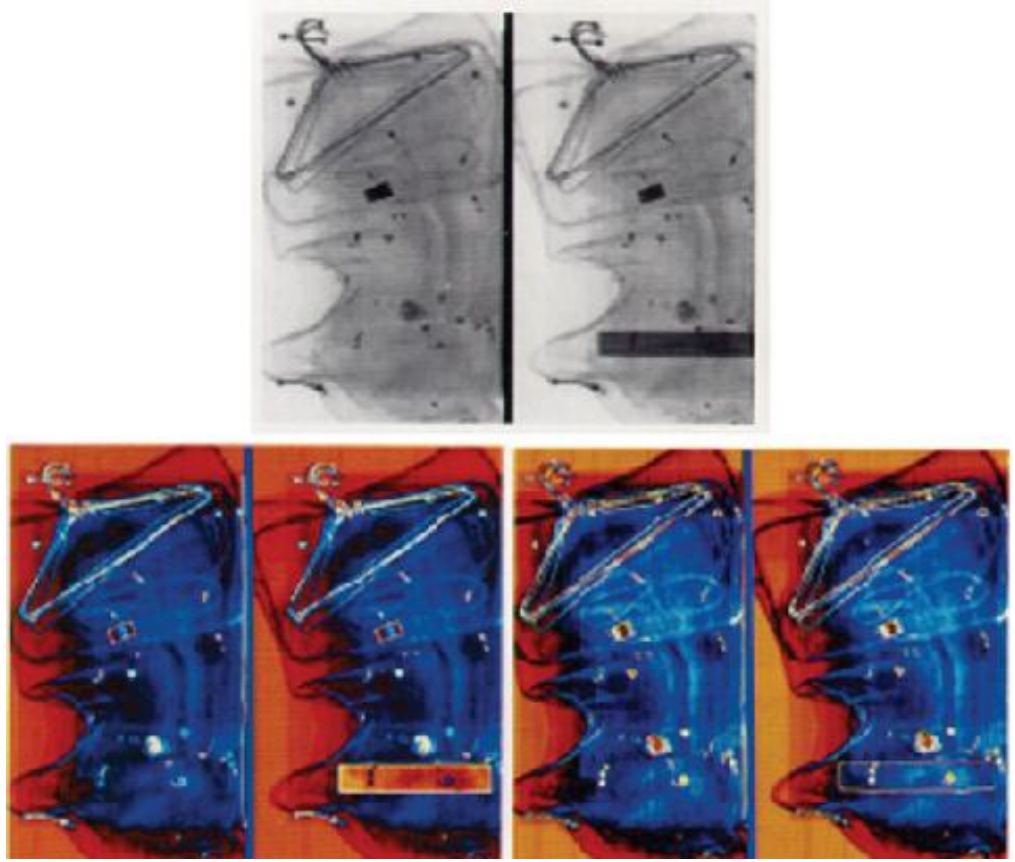


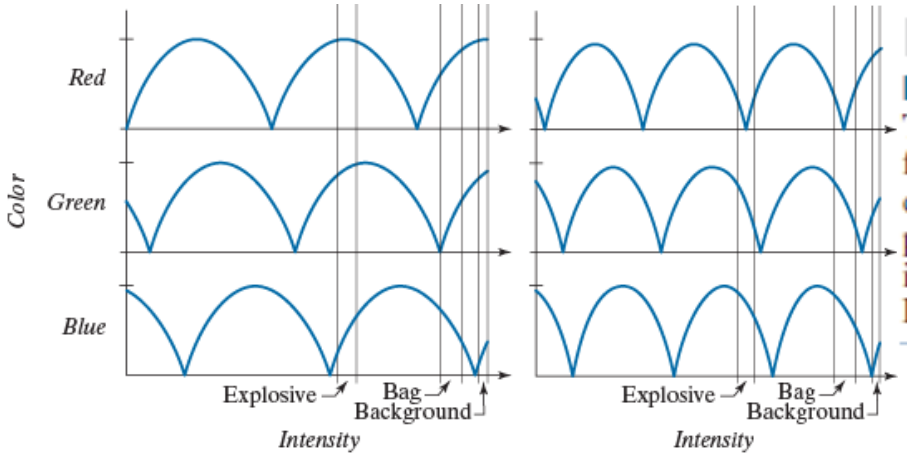
FIGURE 6.21 Functional block diagram for pseudocolor image processing. Images f_R, f_G, and f_B are fed into the corresponding red, green, and blue inputs of an RGB color monitor.

Gray level to Color transformation Application



a
b c

FIGURE 6.22 Pseudocolor enhancement by using the gray level to color transformations in Fig. 6.23. (Original image courtesy of Dr. Mike Hurwitz, Westinghouse.)



a b

FIGURE 6.23 Transformation functions used to obtain the pseudocolor images in Fig. 6.22.

Pseudo-color Image Processing

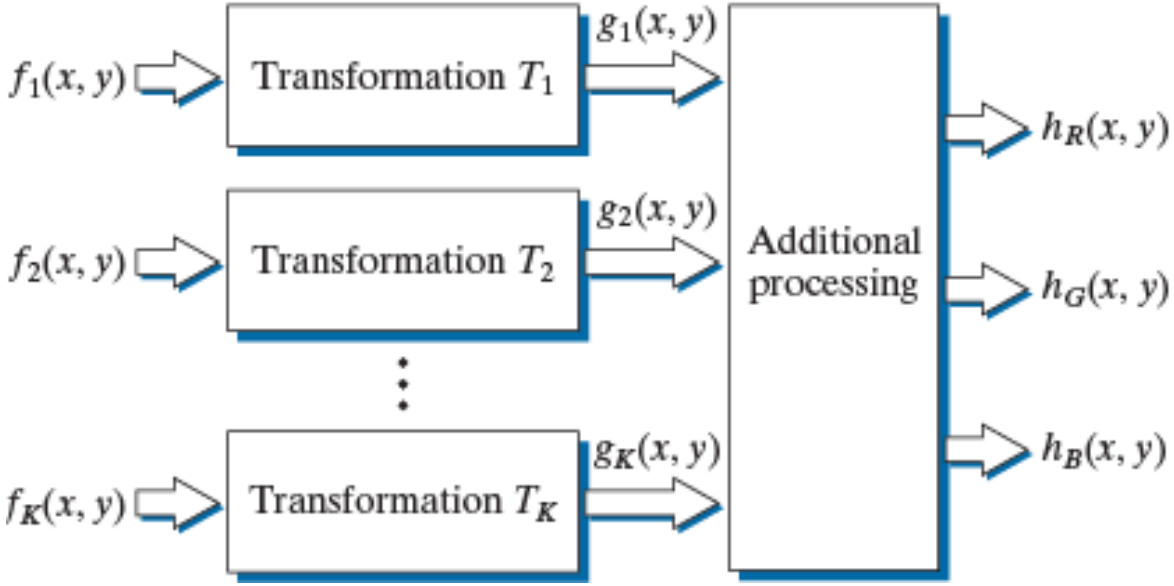
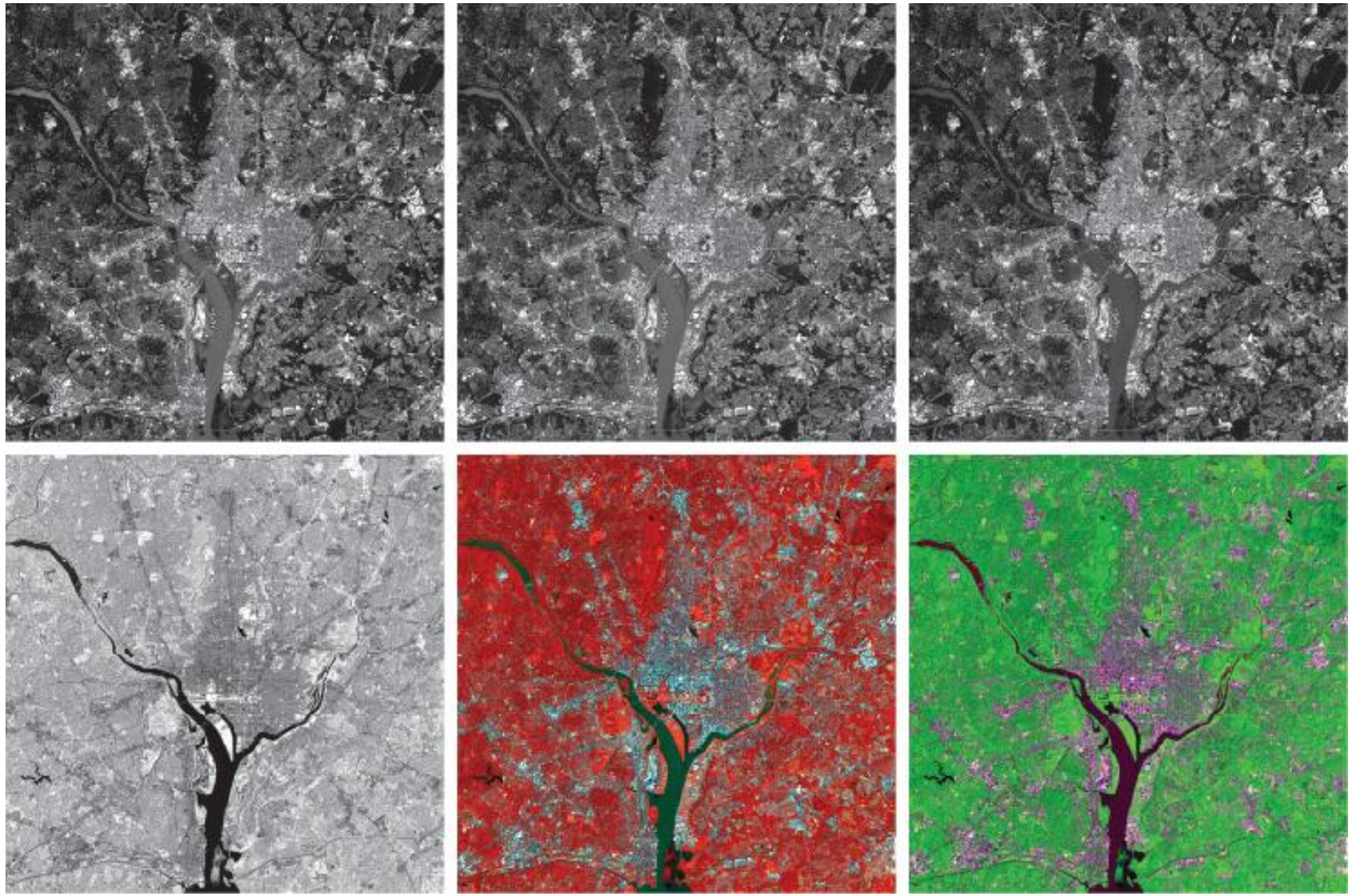


FIGURE 6.24 A pseudocolor coding approach using multiple grayscale images. The inputs are grayscale images. The outputs are the three components of an RGB composite image.

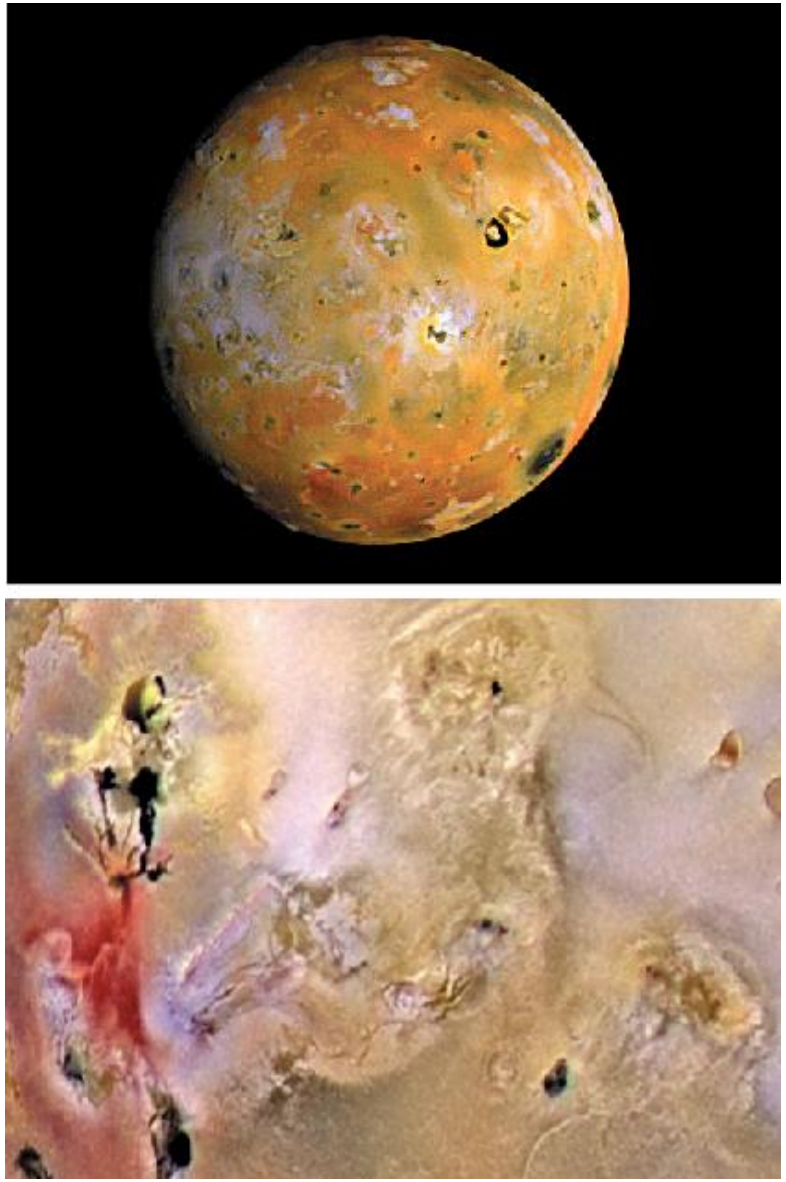
Multi-spectral Imaging



a b c
d e f

FIGURE 6.25 (a)–(d) Red (R), green (G), blue (B), and near-infrared (IR) components of a LANDSAT multispectral image of the Washington, D.C. area. (e) RGB color composite image obtained using the IR, G, and B component images. (f) RGB color composite image obtained using the R, IR, and B component images. (Original multispectral images courtesy of NASA.)

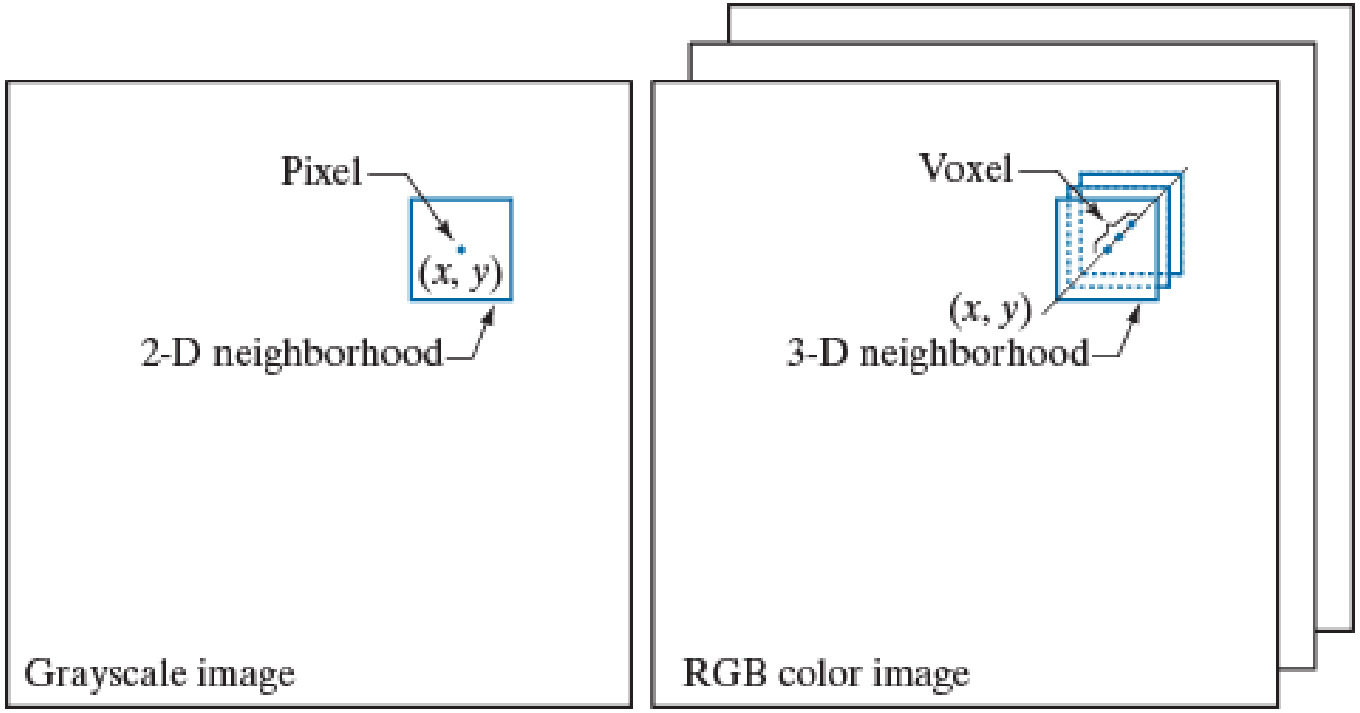
Pseudo-color Image Processing



a
b

FIGURE 6.26 (a) Pseudocolor rendition of Jupiter Moon Io. (b) A close-up. (Courtesy of NASA.)

Full Color Processing



a b

FIGURE 6.27 Spatial neighborhoods for grayscale and RGB color images. Observe in (b) that a *single* pair of spatial coordinates, (x, y), addresses the same spatial location in all three images.

Color Transformation

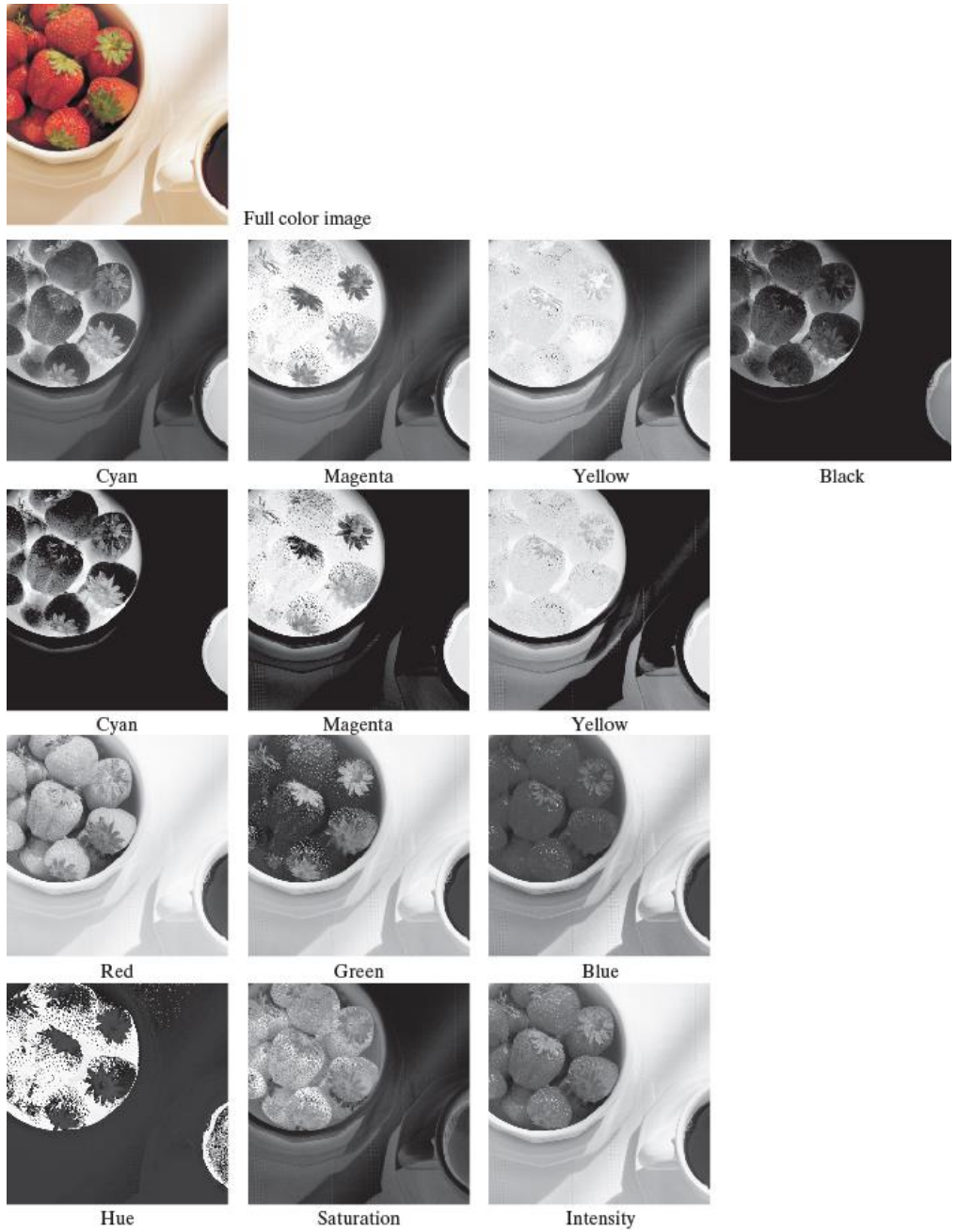


FIGURE 6.28 A full-color image and its various color-space components. (Original image courtesy of MedData Interactive.)

Color Transformation

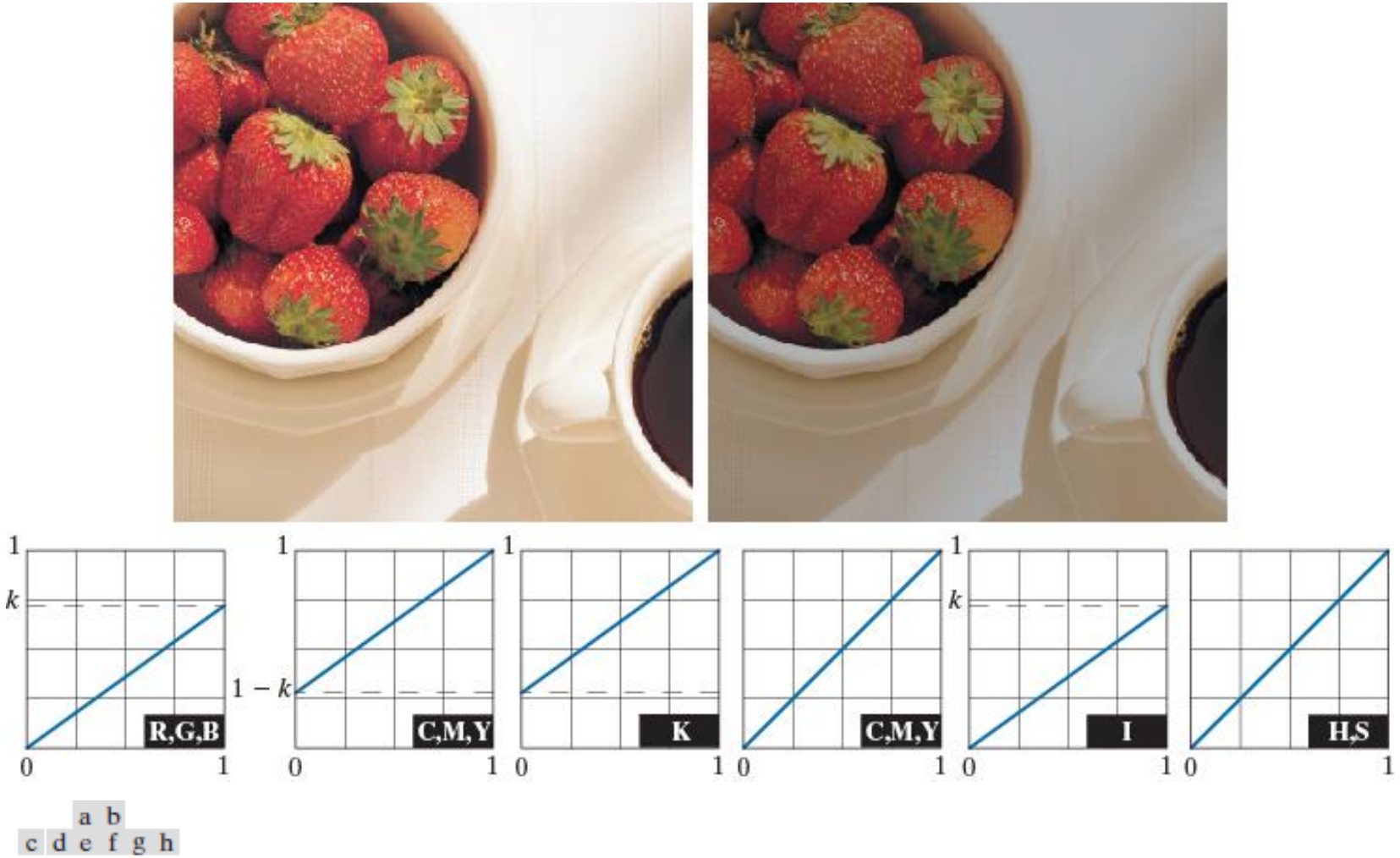


FIGURE 6.29 Adjusting the intensity of an image using color transformations. (a) Original image. (b) Result of decreasing its intensity by 30% (i.e., letting $k = 0.7$). (c) The required RGB mapping function. (d)–(e) The required CMYK mapping functions. (f) The required CMY mapping function. (g)–(h) The required HSI mapping functions. (Original image courtesy of MedData Interactive.)

Color Complements

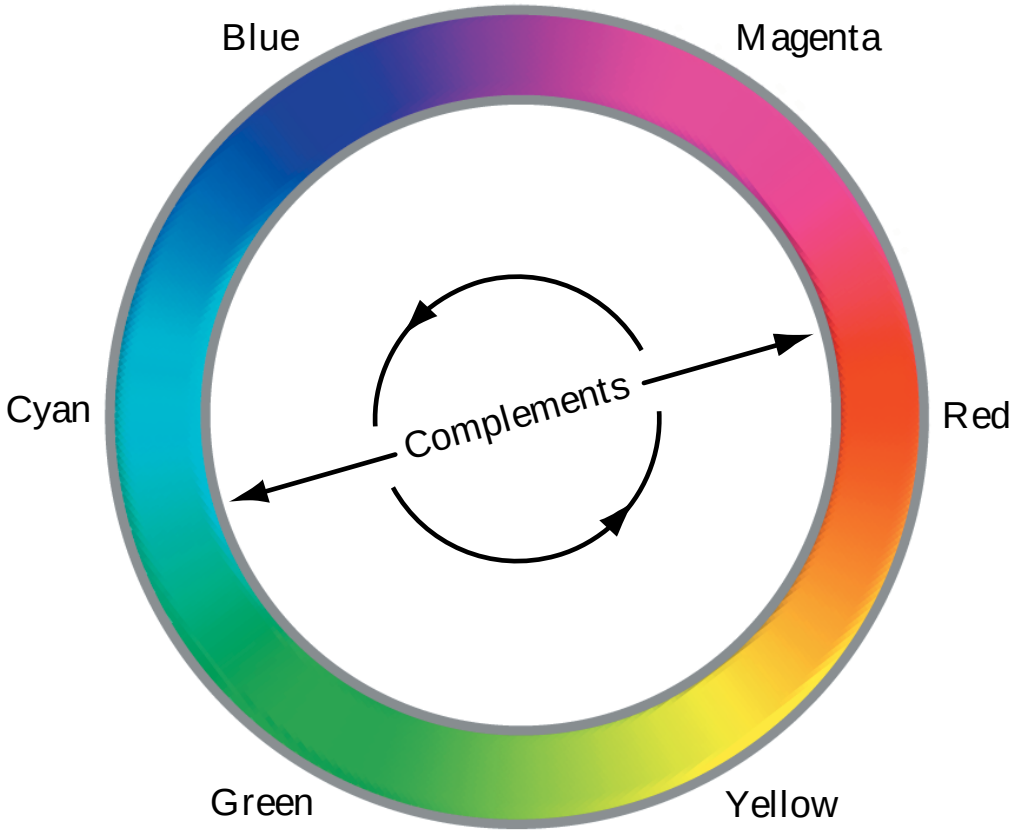
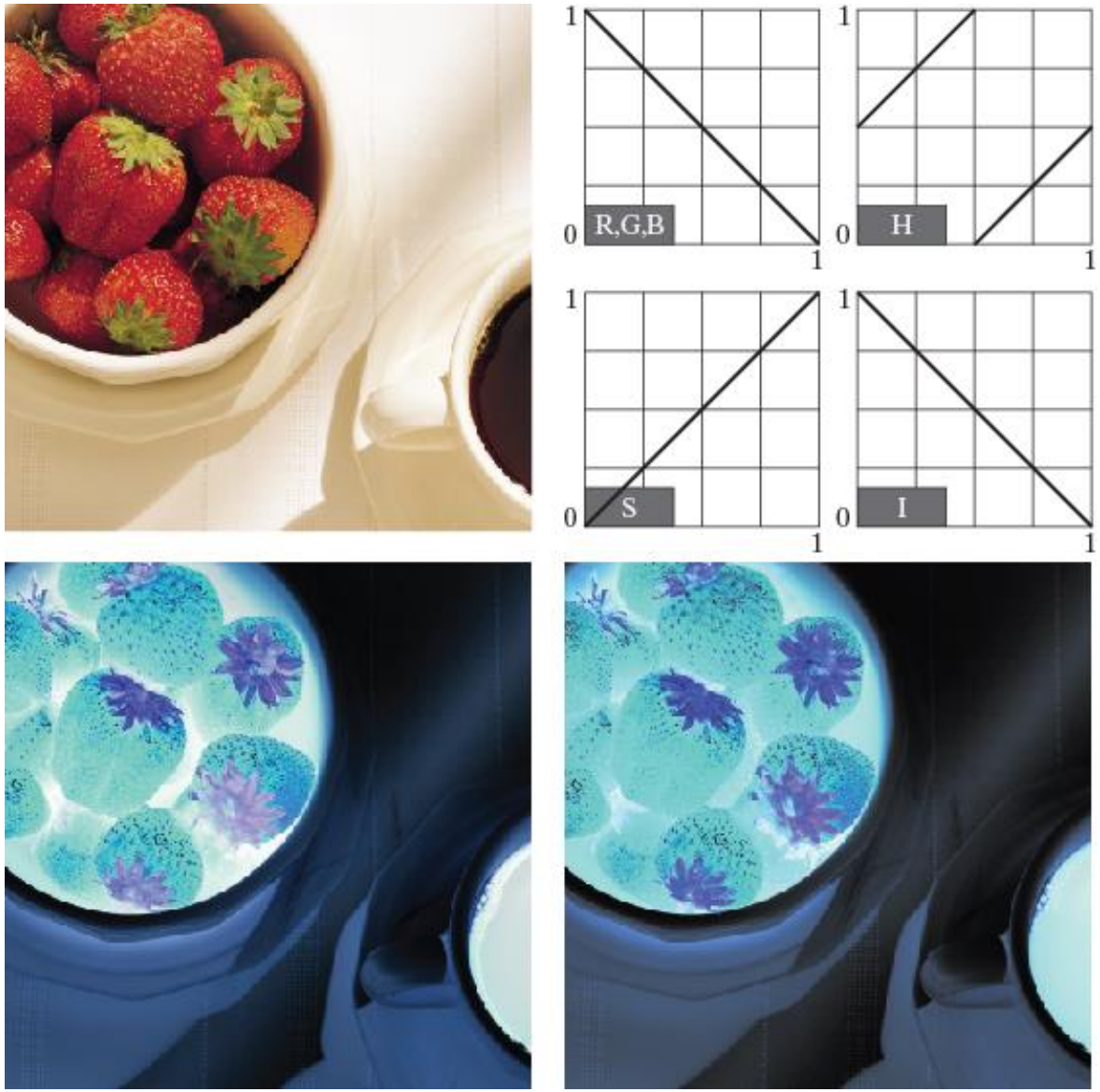


FIGURE 6.30 Color complements on the color circle.

Color Complement Transformation

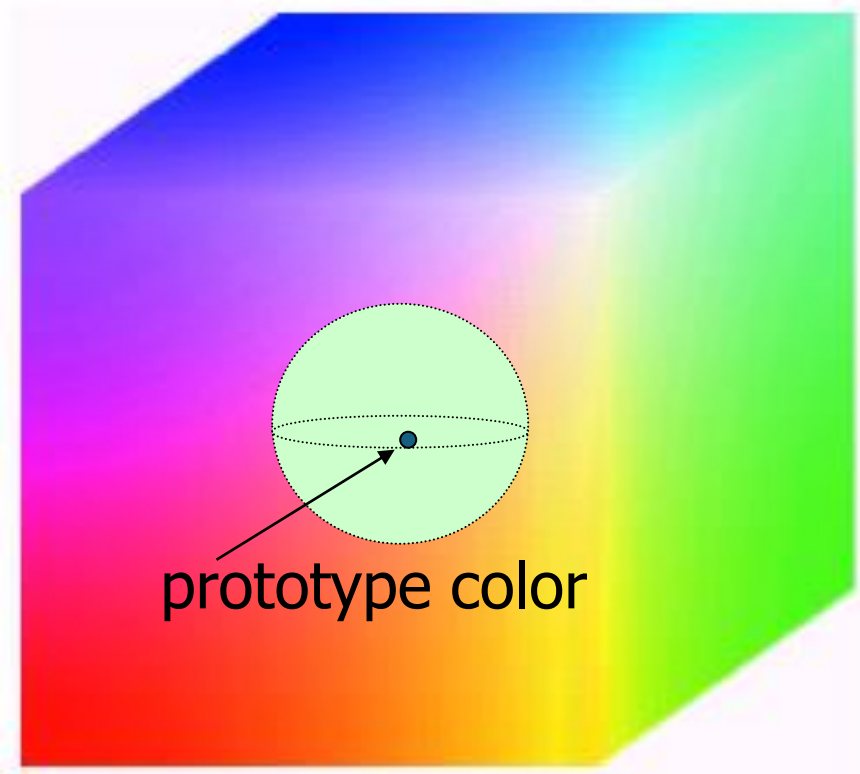


a b
c d

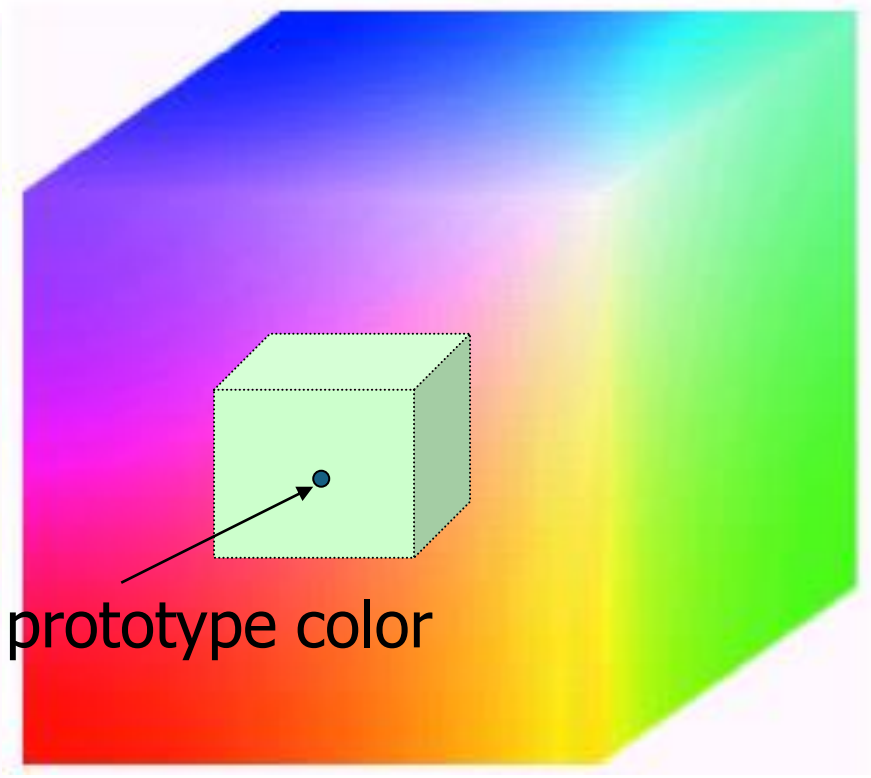
FIGURE 6.31 Color complement transformations. (a) Original image. (b) Complement transformation functions. (c) Complement of (a) based on the RGB mapping functions. (d) An approximation of the RGB complement using HSI transformations.

Color Slicing

- How to take a region of colors of interest?

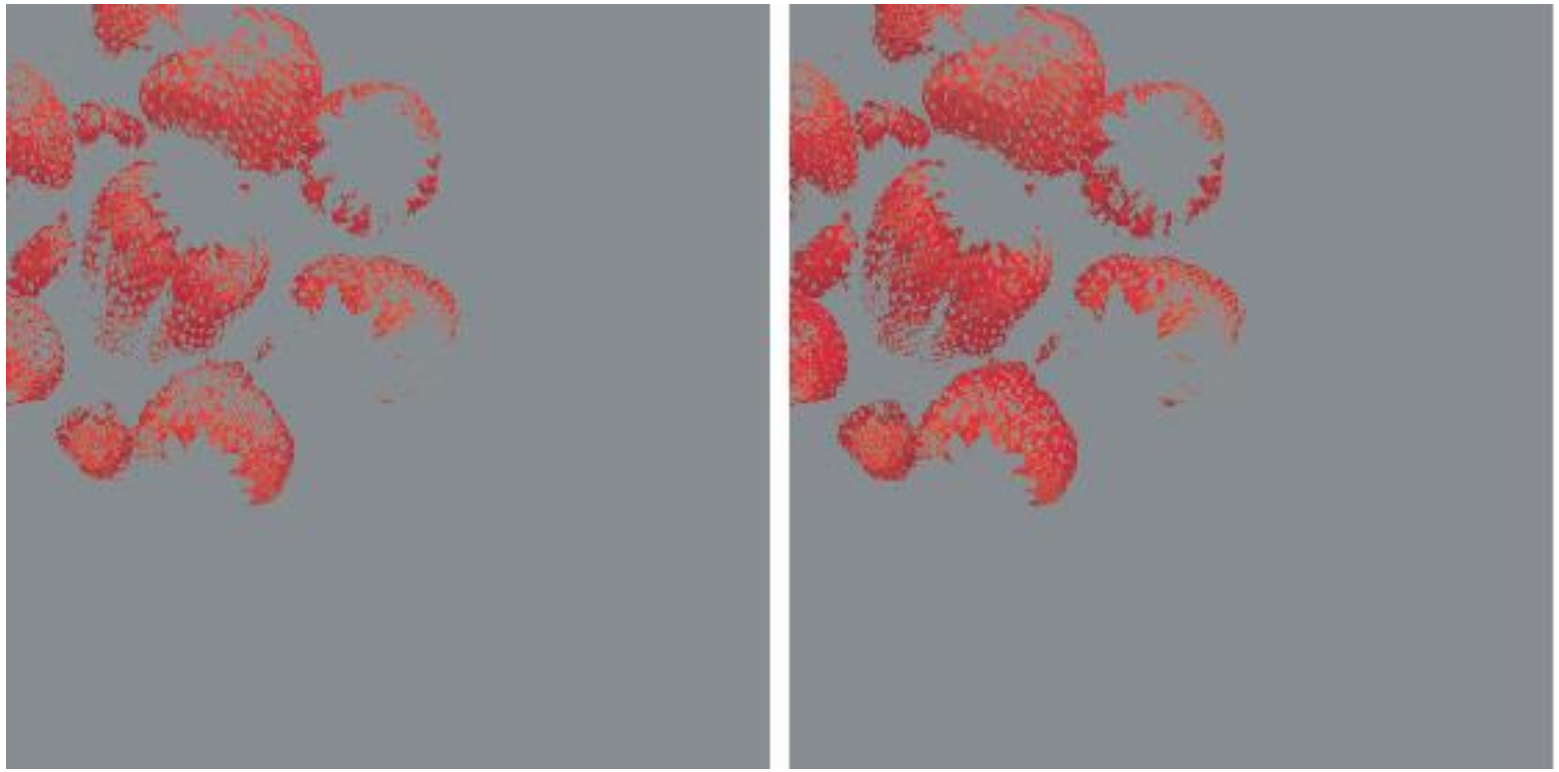


Sphere region



Cube region

Color Slicing Application



a b

FIGURE 6.32 Color-slicing transformations that detect (a) reds within an RGB cube of width $W = 0.2549$ centered at $(0.6863, 0.1608, 0.1922)$, and (b) reds within an RGB sphere of radius 0.1765 centered at the same point. Pixels outside the cube and sphere were replaced by color $(0.5, 0.5, 0.5)$.

Tonal Correction

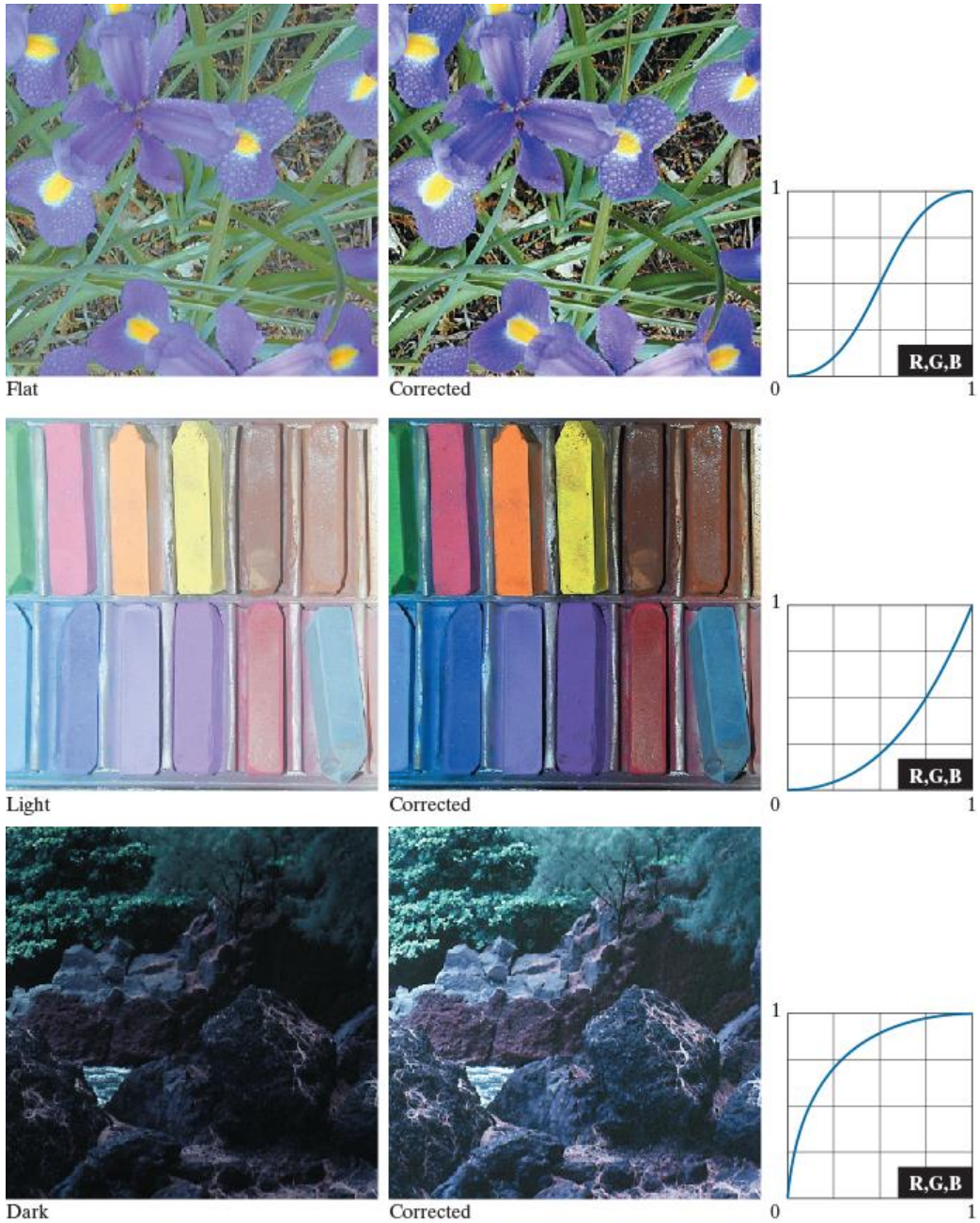


FIGURE 6.33 Tonal corrections for flat, light (high key), and dark (low key) color images. Adjusting the red, green, and blue components equally does not always alter the image hues significantly.

Color Balance

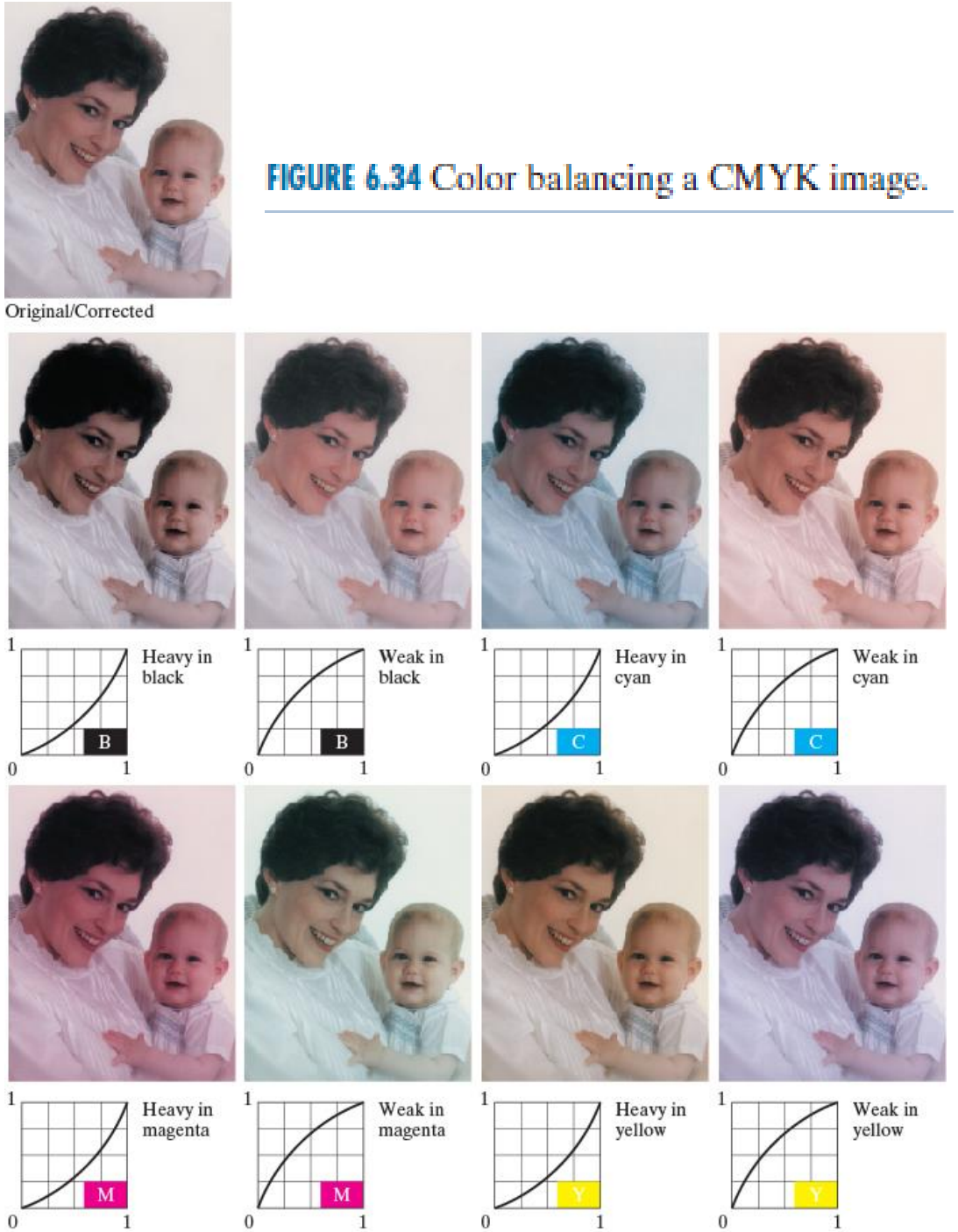
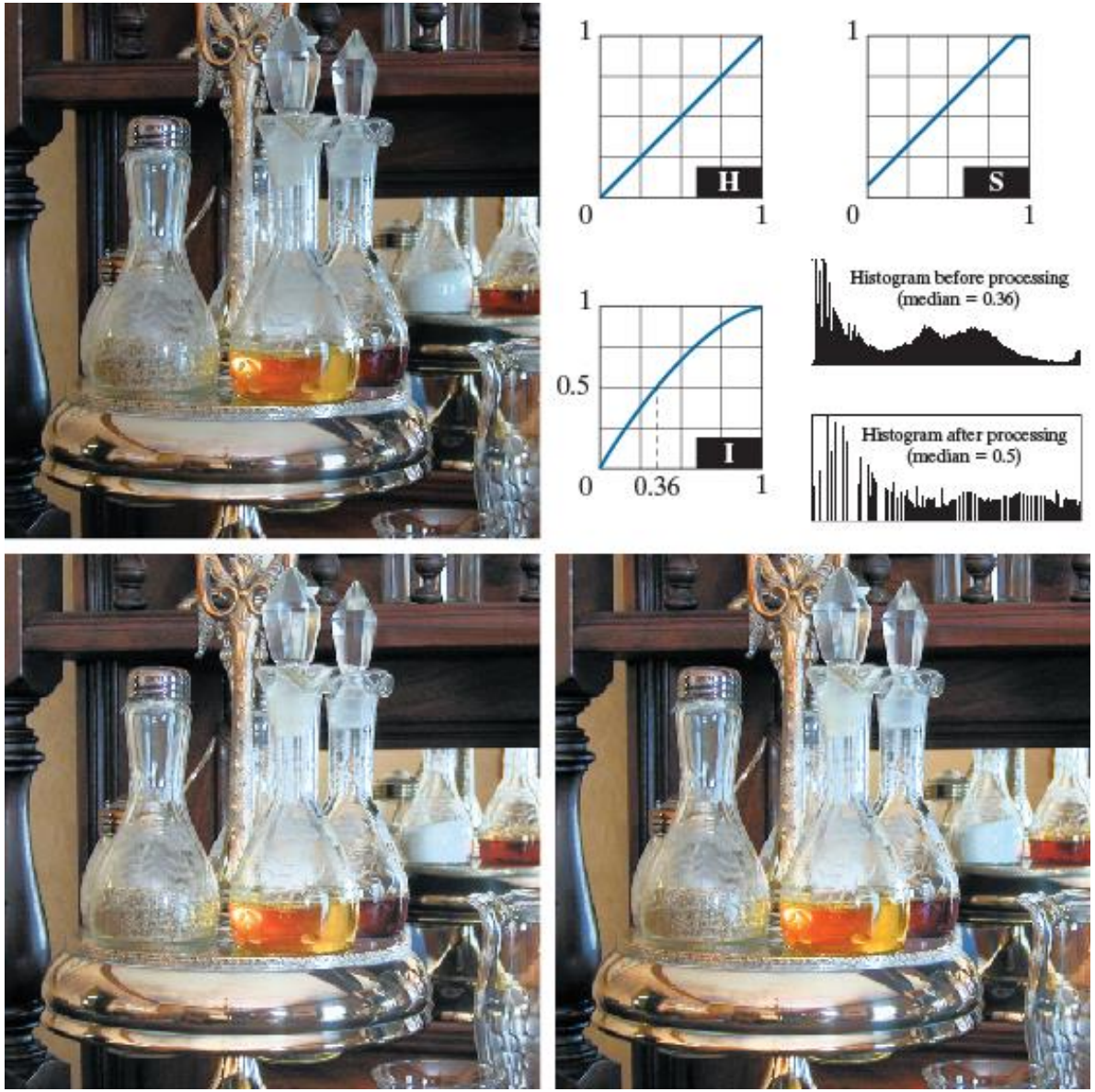


FIGURE 6.34 Color balancing a CMYK image.

Histogram Equalization



a b
c d

FIGURE 6.35 Histogram equalization (followed by saturation adjustment) in the HSI color space.

RGB Color Component Images

a b
c d

FIGURE 6.36
(a) RGB image.
(b) Red component image.
(c)Green component.
(d) Blue component.

HSI Color Component Images

a b c

FIGURE 6.37 HSI components of the RGB color image in Fig. 6.36(a). (a) Hue. (b) Saturation. (c) Intensity.

Image Smoothing

a b c

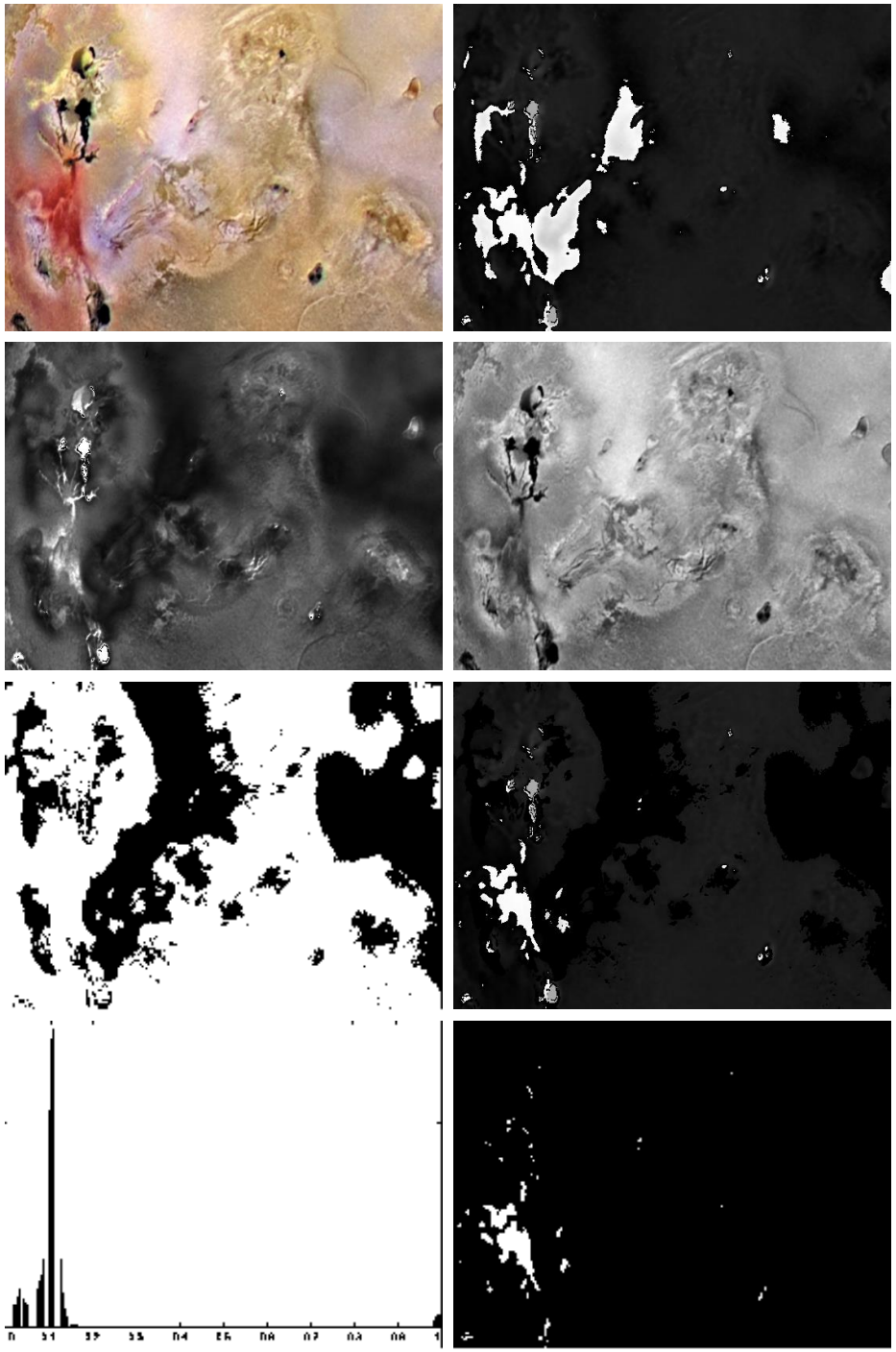
FIGURE 6.38 Image smoothing with a 5×5 averaging kernel. (a) Result of processing each RGB component image. (b) Result of processing the intensity component of the HSI image and converting to RGB. (c) Difference between the two results.

Image Sharpening

a b c

FIGURE 6.39 Image sharpening using the Laplacian. (a) Result of processing each RGB channel. (b) Result of processing the HSI intensity component and converting to RGB. (c) Difference between the two results.

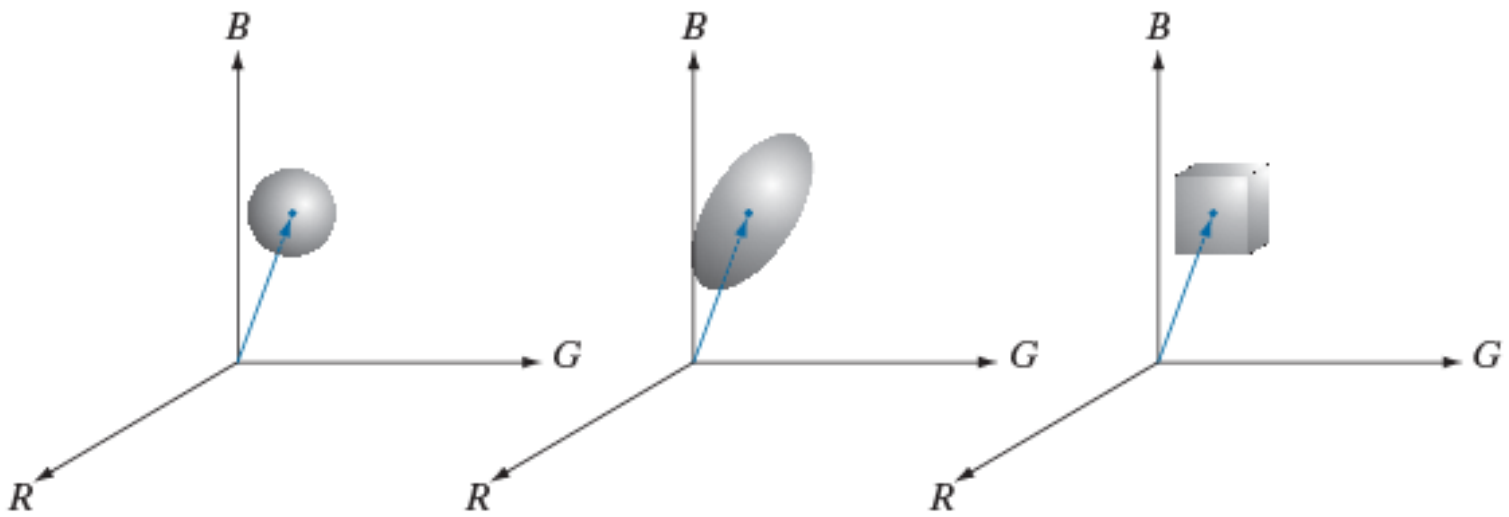
Image Segmentation



a b
c d
e f
g h

FIGURE 6.40 Image segmentation in HSI space. (a) Original. (b) Hue. (c) Saturation. (d) Intensity. (e) Binary saturation mask (black = 0). (f) Product of (b) and (e). (g) Histogram of (f). (h) Segmentation of red components from (a).

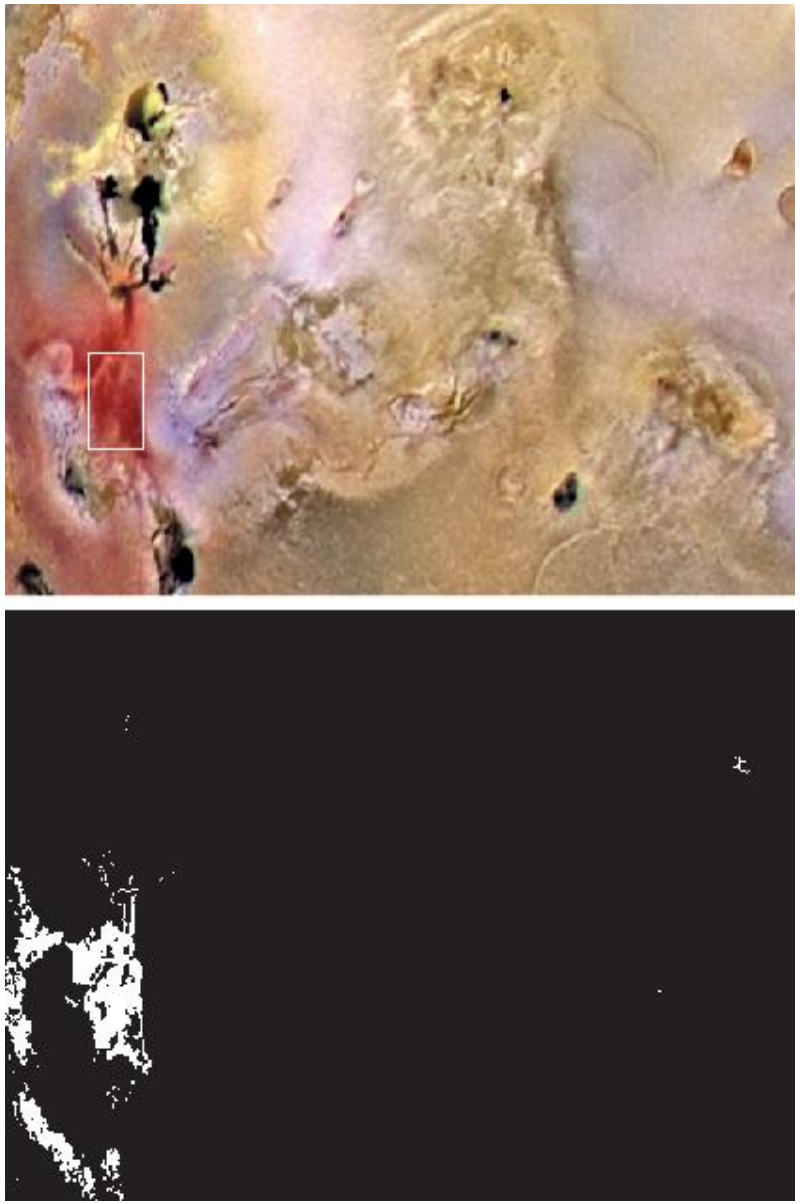
Image Segmentation



a b c

FIGURE 6.41 Three approaches for enclosing data regions for RGB vector segmentation.

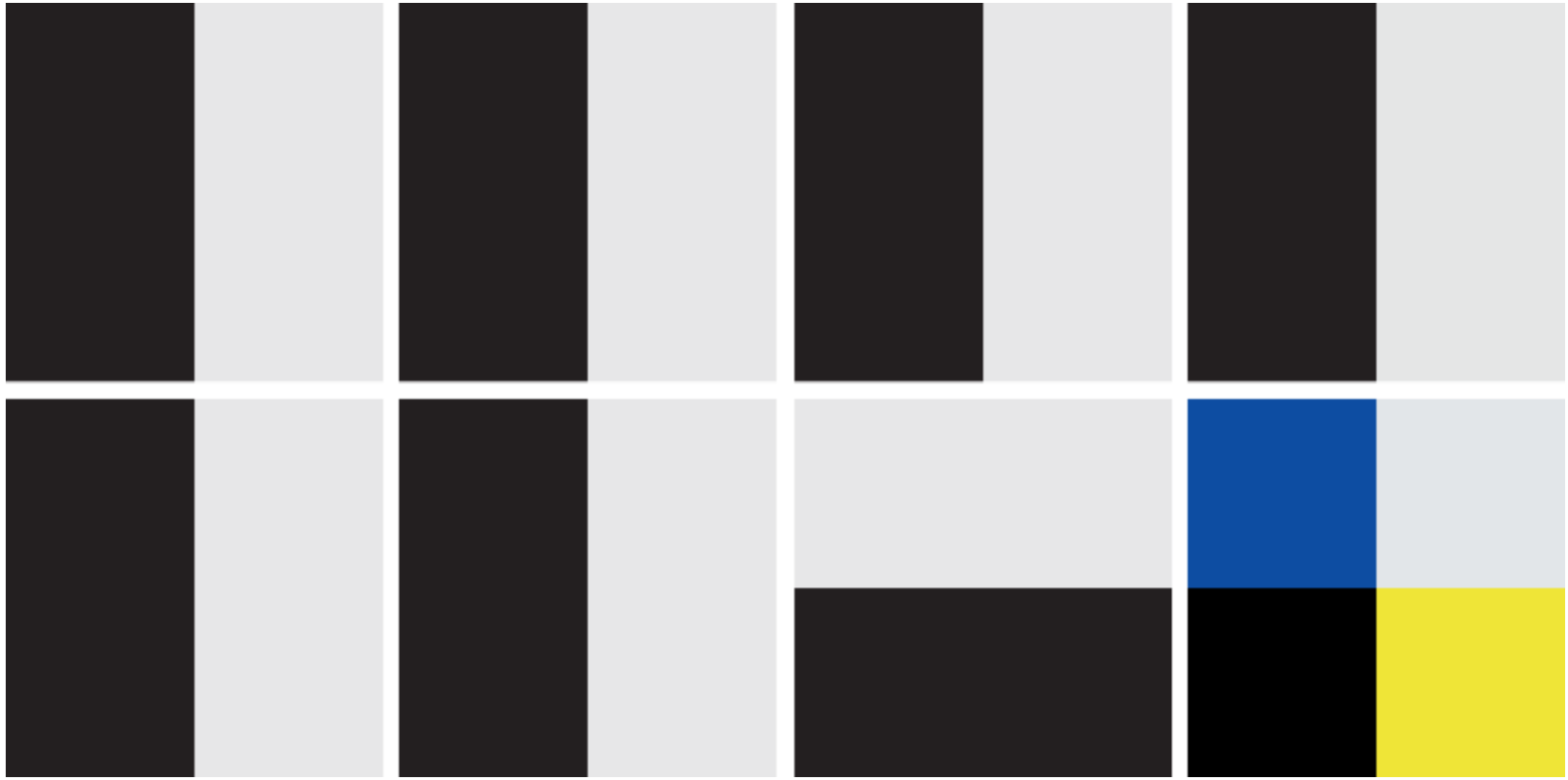
Image Segmentation



a
b

FIGURE 6.42 Segmentation in RGB space. (a) Original image with colors of interest shown enclosed by a rectangle. (b) Result of segmentation in RGB vector space. Compare with Fig. 6.40(h).

RGB Image Components



a b c d
e f g h

FIGURE 6.43 (a)–(c) R, G, and B component images, and (d) resulting RGB color image. (e)–(g) R, G, and B component images, and (h) resulting RGB color image.

Image Gradient

a b
c d

FIGURE 6.44 (a) RGB image. (b) Gradient computed in RGB color vector space. (c) Gradient image formed by the elementwise sum of three individual gradient images, each computed using the Sobel operators. (d) Difference between (b) and (c).

RGB Component Image Gradients

a b c

FIGURE 6.45 Component gradient images of the color image in Fig. 6.44. (a) Red component, (b) green component, and (c) blue component. These three images were added and scaled to produce the image in Fig. 6.44(c).

RGB Noisy Component Image

a b
c d

FIGURE 6.46 (a)–(c) Red, green, and blue 8-bit component images corrupted by additive Gaussian noise of mean 0 and standard deviation of 28 intensity levels. (d) Resulting RGB image. [Compare (d) with Fig. 6.44(a).]

HSI Components of Noisy Color Image

a b c

FIGURE 6.47 HSI components of the noisy color image in Fig. 6.46(d). (a) Hue. (b) Saturation. (c) Intensity.

HSI Components of Noisy Color Image

a b
c d

FIGURE 6.48 (a) RGB image with green plane corrupted by salt-and-pepper noise. (b) Hue component of HSI image. (c) Saturation component. (d) Intensity component.